

SQL Case Study

Sales, Profit & Marketing Analytics

A 29-task SQL assignment analyzing sales, profit, and marketing performance across products and locations, using a Fact-Dimension data model.

- Aggregate Functions — SUM, MIN, MAX, AVG, COUNT
- Multi-table JOINS across Fact, Product & Location tables
- GROUP BY & HAVING for Conditional Aggregation
- Window Functions — DENSE_RANK() for Gap-less Ranking
- CASE Statements for Profit/Loss Classification
- ROLLUP for Hierarchical Weekly Sales Totals
- UNION & INTERSECT Set Operators
- User-Defined Functions & Stored Procedures
- Transactions with ROLLBACK
- String & Math Functions — SUBSTRING, ASCII, BETWEEN
- DELETE Operations with Filtering

Problem Statement:

You are a database administrator. You want to use the data to answer a few questions about your customers, especially about the sales and profit coming from different states, money spent in marketing and various other factors such as COGS (Cost of Goods Sold), budget profit etc. You plan on using these insights to help find out which items are being sold the most. You have been provided with the sample of the overall customer data due to privacy issues. But you hope that these samples are enough for you to write fully functioning SQL queries to help answer the questions.

Dataset:

The 3 key datasets for this case study:

- a. **FactTable:** The Fact Table has 14 columns mentioned below and 4200 rows. Date, ProductID, Profit, Sales, Margin, COGS, Total Expenses, Marketing, Inventory, Budget Profit, Budget COGS, Budget Margin, Budget Sales, and Area Code
Note: COGS stands for Cost of Goods Sold
- b. **ProductTable:** The ProductTable has four columns named Product Type, Product, ProductID, and Type. It has 13 rows which can be broken down into further details to retrieve the information mentioned in the FactTable.
- c. **LocationTable:** Finally, the LocationTable has 156 rows and follows a similar approach to ProductTable. It has four columns named Area Code, State, Market, and Market Size.

Tasks to be performed:

1. Display the number of states present in the LocationTable.
2. How many products are of regular type?
3. How much spending has been done on marketing of product ID 1?
4. What is the minimum sales of a product?
5. Display the max Cost of Good Sold (COGS).
6. Display the details of the product where product type is coffee.
7. Display the details where total expenses are greater than 40.
8. What is the average sales in area code 719?
9. Find out the total profit generated by Colorado state.

Step 1 (contd.): Tasks 10–19

10. Display the average inventory for each product ID.
11. Display state in a sequential order in a Location Table.
12. Display the average budget of the Product where the average budget margin should be greater than 100.
13. What is the total sales done on date 2010-01-01?
14. Display the average total expense of each product ID on an individual date.
15. Display the table with the following attributes such as date, productID, product_type, product, sales, profit, state, area_code.
16. Display the rank without any gap to show the sales wise rank.
17. Find the state wise profit and sales.
18. Find the state wise profit and sales along with the productname.
19. If there is an increase in sales of 5%, calculate the increased sales.

Step 1 (contd.): Tasks 20–29

20. Find the maximum profit along with the product ID and producttype.
21. Create a stored procedure to fetch the result according to the product type from Product Table.
22. Write a query by creating a condition in which if the total expenses is less than 60 then it is a profit or else loss.
23. Give the total weekly sales value with the date and product ID details. Use roll-up to pull the data in hierarchical order.
24. Apply union and intersection operator on the tables which consist of attribute area code.
25. Create a user-defined function for the product table to fetch a particular product type based upon the user's preference.
26. Change the product type from coffee to tea where product ID is 1 and undo it.
27. Display the date, product ID and sales where total expenses are between 100 to 200.
28. Delete the records in the Product Table for regular type.
29. Display the ASCII value of the fifth character from the columnProduct.

Step 2: Dataset Preview — FactTable (fact.csv)

Showing first 10 of 4,248 rows | 14 columns total

Date	ProductId	Profit	Sales	Margin	COGS	Total Expenses	Marketing	Area Code
1/1/2010	1	94	219	130	89	36	24	719
1/1/2010	2	68	190	107	83	39	27	970
1/1/2010	3	101	234	139	95	38	26	970
1/1/2010	13	30	100	56	44	26	14	303
1/1/2010	5	54	134	80	54	26	15	303
1/1/2010	6	53	180	108	72	55	23	720
1/1/2010	8	99	341	171	170	72	47	970
1/1/2010	9	0	150	87	63	87	57	719
1/1/2010	10	33	140	80	60	47	19	970
1/1/2010	11	17	130	72	58	55	22	719

Step 2 (contd.): Dataset Preview — ProductTable (Product.csv)

Showing all 13 rows | 4 columns total

ProductId	Product Type	Product	Type
1	Coffee	Amaretto	Regular
2	Coffee	Columbian	Regular
3	Coffee	Decaf Irish Cream	Decaf
4	Espresso	Caffe Latte	Regular
5	Espresso	Caffe Mocha	Regular
6	Espresso	Decaf Espresso	Decaf
7	Espresso	Regular Espresso	Regular
8	Herbal Tea	Chamomile	Decaf
9	Herbal Tea	Lemon	Decaf
10	Herbal Tea	Mint	Decaf
11	Tea	Darjeeling	Regular
12	Tea	Earl Grey	Regular
13	Tea	Green Tea	Regular

Step 2 (contd.): Dataset Preview — LocationTable (Location.csv)

Showing first 10 of 156 rows | 4 columns total

Area Code	State	Market	Market Size
203	Connecticut	East	Small Market
206	Washington	West	Small Market
209	California	West	Major Market
210	Texas	South	Major Market
212	New York	East	Major Market
213	California	West	Major Market
214	Texas	South	Major Market
216	Ohio	Central	Major Market
217	Illinois	Central	Major Market
224	Illinois	Central	Major Market

Step 2: Setup + Tasks 1 & 2 — State Count, Regular Product Count

```
SQL Mandator...J\DELL (86))
J...V\DELL (89))  X  tzmsnpfg...sql...J\DELL (87))

CREATE DATABASE SalesDB;
GO

USE SalesDB;
GO

SELECT TOP 10 *
FROM dbo.fact;

select top 10 *
from dbo.location;

select top 10 *
from dbo.Product;

--1. Display the number of states present in LocationTable
SELECT COUNT(DISTINCT State) AS NumberOfStates
FROM dbo.location;

--2. How many products are of Regular type?
SELECT COUNT(*) AS RegularProducts
FROM dbo.Product
WHERE Type = 'Regular';
```

Step 3: Tasks 3–6 — Marketing Spend, Min Sales, Max COGS, Coffee Products

--3.How much spending has been done on marketing of Product ID 1?

```
SELECT SUM(Marketing) AS TotalMarketing
FROM dbo.Fact
WHERE ProductID = 1;
```

--4.What is the minimum sales of a product?

```
SELECT MIN(Sales) AS MinimumSales
FROM dbo.Fact;
```

--5.Display the maximum COGS

```
SELECT MAX(COGS) AS MaximumCOGS
FROM dbo.Fact;
```

--6.Display the details of the product where Product Type is Coffee

```
SELECT TOP 5 *
FROM dbo.Product;
```

```
SELECT *
FROM dbo.Product
WHERE Product_Type = 'Coffee';
```

--7.Display the details where Total Expenses are greater than 40

```
SELECT *  
FROM dbo.Fact  
WHERE [Total_Expenses] > 40;
```

--8.What is the average sales in Area Code 719?

```
SELECT AVG(Sales) AS AverageSales  
FROM dbo.Fact  
WHERE [Area_Code] = 719;
```

--9.Find the total profit generated by Colorado state

```
SELECT SUM(F.Profit) AS TotalProfit  
FROM dbo.Fact F  
JOIN dbo.Location L  
    ON F.[Area_Code] = L.Area_Code  
WHERE L.State = 'Colorado';
```

--10. Display the average inventory for each Product ID

```
SELECT  
    ProductID,  
    AVG(Inventory) AS AverageInventory  
FROM dbo.Fact  
GROUP BY ProductID;
```

Step 5: Tasks 11–13 — Sequential States, Budget Filter (HAVING), Total Sales by Date

--11.Display state in a sequential order in Location Table

```
SELECT *  
FROM dbo.Location  
ORDER BY State ASC;
```

--12.Display the average budget of the Product where
--the average budget margin is greater than 100

```
SELECT  
    P.Product,  
    AVG(F.[Budget_Sales]) AS AverageBudget  
FROM dbo.Product P  
JOIN dbo.Fact F  
    ON P.ProductID = F.ProductID  
GROUP BY P.Product  
HAVING AVG(F.[Budget_Margin]) > 100;
```

--13.What is the total sales done on date 2010-01-01?

```
SELECT  
    SUM(Sales) AS TotalSales  
FROM dbo.Fact  
WHERE [Date] = '2010-01-01';
```

Step 6: Tasks 14–15 — Avg Expense per Product/Date, Full Joined Table

--14.Display the average total expense of each Product ID on an individual date

```
SELECT
    [Date],
    ProductID,
    AVG([Total_Expenses]) AS AverageTotalExpense
FROM dbo.Fact
GROUP BY [Date], ProductID
ORDER BY [Date], ProductID;
```

--15.Display Date, ProductID, Product Type, Product, Sales, Profit,
--State, Area Code

```
SELECT
    F.[Date],
    F.ProductID,
    P.[Product_Type],
    P.Product,
    F.Sales,
    F.Profit,
    L.State,
    F.[Area_Code]
FROM dbo.Fact F
JOIN dbo.Product P
    ON F.ProductID = P.ProductID
JOIN dbo.Location L
    ON F.[Area_Code] = L.Area_Code;
```

Step 7: Tasks 16–17 — DENSE_RANK Sales Ranking, State-wise Profit & Sales

--16.Display the rank without any gap to show the sales-wise rank

```
SELECT
    ProductID,
    Sales,
    DENSE_RANK() OVER (ORDER BY Sales DESC) AS SalesRank
FROM dbo.Fact;
--17.Find the state-wise profit and sales
```

```
SELECT
    L.State,
    SUM(F.Profit) AS TotalProfit,
    SUM(F.Sales) AS TotalSales
FROM dbo.Fact F
JOIN dbo.Location L
    ON F.[Area_Code] = L.Area_Code
GROUP BY L.State;
```

Step 8: Tasks 18–19 — Profit/Sales with Product Name, 5% Sales Increase

--18.Find state-wise profit and sales along with Product Name

```
SELECT
    L.State,
    P.Product,
    SUM(F.Profit) AS TotalProfit,
    SUM(F.Sales) AS TotalSales
FROM dbo.Fact F
JOIN dbo.Location L
    ON F.[Area_Code] = L.Area_Code
JOIN dbo.Product P
    ON F.ProductID = P.ProductID
GROUP BY L.State, P.Product;
```

--19.If there is an increase in sales of 5%, calculate the increased sales

```
SELECT
    ProductID,
    Sales,
    Sales * 1.05 AS IncreasedSales
FROM dbo.Fact;
```

Step 9: Tasks 20–21 — Max Profit Lookup, Stored Procedure by Product Type

--20.Find the maximum profit along with Product ID and Product Type

```
SELECT
    F.ProductID,
    P.[Product_Type],
    F.Profit
FROM dbo.Fact F
JOIN dbo.Product P
    ON F.ProductID = P.ProductID
WHERE F.Profit = (
    SELECT MAX(Profit)
    FROM dbo.Fact
);
```

--21.Create a stored procedure to fetch the result according to Product Type

```
CREATE OR ALTER PROCEDURE GetProductDetails
    @ProductType VARCHAR(50)
AS
BEGIN
    SELECT *
    FROM dbo.Product
    WHERE [Product_Type] = @ProductType;
END;
GO

EXEC GetProductDetails 'Coffee';
```

Step 10: Tasks 22–24 — Profit/Loss CASE Logic, Weekly ROLLUP, UNION & INTERSECT

SQL Mandator...JV\DELL (86))

--22.If Total Expenses are less than 60 → Profit, otherwise → Loss

```
SELECT
    ProductID,
    [Total_Expenses],
    CASE
        WHEN [Total_Expenses] < 60 THEN 'Profit'
        ELSE 'Loss'
    END AS Status
FROM dbo.Fact;
```

--23 Total weekly sales with Date and ProductID using ROLLUP

```
SELECT
    DATEPART(WEEK, [Date]) AS WeekNumber,
    ProductID,
    SUM(Sales) AS TotalWeeklySales
FROM dbo.Fact
GROUP BY ROLLUP(DATEPART(WEEK, [Date]), ProductID);
```

--24 Apply UNION and INTERSECT on tables containing Area Code

```
SELECT [Area_Code] FROM dbo.Fact
UNION
SELECT Area_Code FROM dbo.Location;
```

```
SELECT [Area_Code] FROM dbo.Fact
INTERSECT
SELECT Area_Code FROM dbo.Location;
```

Step 11: Task 25 — User-Defined Function for Product Type Lookup

--25 Create a user-defined function to fetch Product
--Type based on user's preference

```
CREATE FUNCTION GetProduct_Type
(
    @ProductID INT
)
RETURNS VARCHAR(50)
AS
BEGIN
    DECLARE @ProductType VARCHAR(50);

    SELECT @ProductType = [Product_Type]
    FROM dbo.Product
    WHERE ProductID = @ProductID;

    RETURN @ProductType;
END;
GO

SELECT dbo.GetProduct_Type(1) AS ProductType;
```

Step 12: Tasks 26–27 — Transaction Update with ROLLBACK, Expense Range Filter

```
|
--26 Change Product Type from Coffee to Tea where ProductID = 1 and undo it

BEGIN TRANSACTION;

UPDATE dbo.Product
SET [Product_Type] = 'Tea'
WHERE ProductID = 1
AND [Product_Type] = 'Coffee';

SELECT *
FROM dbo.Product
WHERE ProductID = 1;

ROLLBACK TRANSACTION;

--27.Display Date, ProductID and Sales where Total Expenses are between 100 and 200

SELECT
    [Date],
    ProductID,
    Sales
FROM dbo.Fact
WHERE [Total_Expenses] BETWEEN 100 AND 200;
```

Step 13: Tasks 28–29 — DELETE Regular Products, ASCII of 5th Character

--28.Delete records in ProductTable for Regular type

```
DELETE FROM dbo.Product  
WHERE Type = 'Regular';
```

```
SELECT * FROM dbo.Product  
WHERE Type = 'Regular';
```

--29. Display ASCII value of the fifth character from Product

```
SELECT  
    Product,  
    SUBSTRING(Product, 5, 1) AS FifthCharacter,  
    ASCII(SUBSTRING(Product, 5, 1)) AS ASCIIValue  
FROM dbo.Product;
```